

The Elimination Ledger

A Catalogue of Falsifications, Refractions, and Survivals
in the ACS / AISO Program

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June 2026 | SIP License v1.1

Abstract

This paper reports negative results as first-class outputs. Across a single elimination campaign we subjected the load-bearing claims of the ACS (Asymmetric Codependent Systems) framework and the AISO/TENT computational stack to pre-registered kill tests, each carrying an explicit falsification criterion fixed *before* the test was run. We adopt the discipline of *adversarial compression*: a claim is admitted as a theorem only after it survives an attempt to destroy it, and a claim that fails is recorded — with its killing computation — as a result of equal standing. The campaign retired the program’s two highest-collapse claims, both of which were the program’s own: the information-asymmetry / central-charge identity $\Delta I \equiv c$, and the conditional-uniformity height floor $T_{\min} = (2\pi e)^d/q$. Several numerical constants long carried as “forced” were shown to be *refractions* — artifacts of a normalization or counting convention — while the *relations* among them survived. We also gather the program’s accumulated dead ends from prior work into a single catalogue. The recurring lesson is that the quantities which survive a change of instrument are relations, ratios, and equalities; bare convention-laden numbers are not. Every verdict is labelled by a four-tier verification hierarchy, and every kill test states what it does and does not claim.

Publication status (24 September 2026). This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

Contents

1	Introduction: overclaiming as the primary failure mode	2
2	Methodology	2
2.1	Adversarial compression	2
2.2	The four-tier verification hierarchy	3
2.3	The tomographic invariance engine and its kill-criterion	3
2.4	Pre-registration, controls, decoys	3
3	Results I: the invariant/refraction classification of framework constants	3

4	Results II: falsifications	4
4.1	The height-floor scaling law $T_{\min} = (2\pi e)^d/q$ (T4)	4
4.2	The information-asymmetry / central-charge identity $\Delta I \equiv c$ (T4 as stated)	5
4.3	The BRA speed-superiority claim (T4 for speed; determinism survives at T1)	6
4.4	Residual prime-orbit off-diagonal mechanisms (T2 + T3)	6
5	Results III: class-level eliminations — the Hilbert–Pólya operator	7
6	Results IV: a tension resolved by decoupling (T3 + T2)	7
7	The accumulated dead ends	8
8	Discussion: recurring patterns	8
9	What survives, and what remains open	9
10	Conclusion	9
A	Reproducibility	9

1 Introduction: overclaiming as the primary failure mode

In a long-running theoretical program the dominant risk is not error but *overclaiming*: dressing a conjecture in the language of a theorem, or carrying a convention-dependent number as though it were a forced prediction. The remedy adopted here is to make destruction cheap and routine. Rather than accumulate positive support for favored claims, we attempt to kill them, and we treat a clean kill as a success. This is the strip-mining stance: one does not prove the gold is present; one removes everything that is not gold, and what remains earns credibility by the elimination of alternatives.

The campaign reported here ran a queue of targets, each a load-bearing claim of the ACS framework or the AISO/TENT stack, and each equipped with a pre-registered kill condition. The targets were ordered by expected collapse of candidate space per unit cost, with deliberate weight given to claims that were *our own*. A kill aimed at one’s own load-bearing claim is the highest-integrity move available, and — as the results below show — the most productive: the two largest results of the campaign are both falsifications of claims the program had previously advanced.

2 Methodology

2.1 Adversarial compression

Every conjecture receives a pre-registered test with a kill condition.

- A survivor becomes a theorem only after a proof is supplied.
- A failure is retained as a negative result with the killing computation.
- A partial result remains an observation with an explicit scope boundary.

The rules are non-negotiable: a conjecture is never described in theorem language until its proof is complete; a negative result is documented with its conjecture, its killing computation, the mechanism of failure, and the boundary it establishes; every scope boundary states what the result does and does not claim.

2.2 The four-tier verification hierarchy

Each verdict is labelled by the strongest tier it has earned, and a lower tier is never allowed to pass as a higher one.

Tier	Standard	Example in this paper
T1	machine-measured, reproducible	g_4 moves $4/3 \rightarrow 2/3$ under rescaling
T2	proven, or true by inspection	Λ supported on prime powers
T3	numerical, not yet theorem-level	GUE level-repulsion $\beta \approx 2.12$
T4	explicitly falsified by computation	$T_{\min} = (2\pi e)^d/q$ at $d = 2$

2.3 The tomographic invariance engine and its kill-criterion

A central instrument is a *tomographic invariance engine*: a battery that varies the measuring instrument (representation, normalization, convention, scale, analysis window, taper) and asks which quantities move. The operative definition:

A quantity is a **refraction** if its value moves under a legitimate change of instrument.
It is an **invariant** (candidate) if it does not.

The criterion is applied either numerically — by recomputing under an instrument swap, yielding a T1 verdict — or structurally, by argument from a quantity’s construction when no recomputable pipeline is reachable, yielding a T2 verdict. Before trusting the engine on genuinely uncertain calls, it was calibrated against known invariants and known refractions, with negative controls and decoy targets included so that the engine could not become its own refraction.

2.4 Pre-registration, controls, decoys

Each kill test fixes its falsification criterion before execution. Negative controls (quantities that should show nothing) and decoys (near-miss targets that should not register) accompany every numerical test. Two further disciplines were enforced throughout: *measured, never projected* — a complexity argument or extrapolation is not a measurement and is never reported as one — and *validate the instrument before trusting it*, e.g. a new numerical method must reproduce an independent method on a shared case before it is used at scale.

3 Results I: the invariant/refraction classification of framework constants

The ACS parameter ledger carried four quantities as algebraically “locked.” We turned the engine inward and recomputed each under a legitimate change of representation or counting convention. The verdicts (all T1, machine-confirmed by re-running the derivations under instrument swaps) are summarized in Table 1.

Quantity	As carried	Under instrument swap	Verdict
$g_4 = g_L = g_R$	4/3	$\rightarrow 2/3$ under $\text{Tr}(T^a T^b) = \frac{1}{2}\delta$ vs δ	refraction (value)
γ (Barbero–Immirzi)	0.274067	$\rightarrow 0.190206$ under $\text{SU}(2) \rightarrow \text{SO}(3)$ counting	refraction
λ_ϕ (Higgs quartic)	$2\sqrt{3}/27 \approx 0.1283$	scales with Killing-form norm.	refraction
$\tilde{h}/h$ (Yukawa ratio)	2/3	$d/d(\ln \mu) \approx 0$, normalization cancels	invariant

Table 1: Framework constants under the tomographic kill-criterion. Three bare values are refractions; one dimensionless ratio is an invariant.

$g_4 = 4/3$. A bare gauge-coupling *value* is generator-normalization dependent; rescaling the trace form $\text{Tr}(T^a T^b) = k \delta^{ab}$ sends $4/3 \rightarrow 2/3$. The *equality* $g_4 = g_L = g_R$, however, is an equality of three couplings in one shared normalization and is preserved under any consistent rescaling. The value is a refraction; the unification relation is the invariant.

$\gamma = 0.274$. The value is counting-prescription dependent. Under the $\text{SU}(2)$ (half-integer) counting it is 0.274067; under $\text{SO}(3)$ (integer) counting it is 0.190206, a shift of $|\Delta\gamma| = 0.0839$. The value moves under a change of the state-counting prescription; only the existence of a fixed γ *given* a prescription is structural.

$\lambda_\phi = 2\sqrt{3}/27$. The numerical value scales with the generator/Killing-form normalization. The Koide-geometric *origin* of the quartic is the candidate invariant; the specific number, and its identification with the physical running quartic (which crosses experiment only near $\mu \sim 132$ GeV), are scale- and convention-relative.

$\tilde{h}/h = 2/3$. A dimensionless ratio in which multiplicative normalization cancels, additionally protected as a renormalization-group invariant ($d/d(\ln \mu)(\tilde{h}/h) \approx 0$ on integration of the Yukawa system). This is the strongest invariant candidate of the four and the only one that survives.

The pattern. Across all four, the relations survive and the bare numbers do not: the equality $g_4 = g_L = g_R$, the ratio $\tilde{h}/h$, and the geometric origin of λ_ϕ are invariant or candidate-invariant; the convention- and scale-laden numbers 4/3, 0.274, 0.1283 are refractions. This is the expected signature of physics, in which observables are relations.

4 Results II: falsifications

4.1 The height-floor scaling law $T_{\min} = (2\pi e)^d/q$ (T4)

The program’s conditional-uniformity height floor was stated as $T_{\min}(d, q) = (2\pi e)^d/q$, with the mechanism that $T_{\min}$ is the height at which the Riemann–von Mangoldt count

$$N(T) \sim \frac{dT}{2\pi} \log \frac{qT}{(2\pi e)^d} \quad (1)$$

becomes asymptotically valid, i.e. $qT > (2\pi e)^d$, below which “too few zeros exist to resolve the d -fold denser prime spectrum.”

For the Riemann zeta function ($d = 1, q = 1$) the floor is $2\pi e \approx 17.08$, exactly the zero of the leading count term $(T/2\pi)(\log(T/2\pi) - 1)$ — a window-free analytic identity, with decoy heights

$(2\pi e \cdot \varphi, (2\pi e)^2/10)$ not privileged. At $d = 1$ this floor is a genuine invariant. But $d = 1$ is the degenerate case: the standard analytic-conductor zero count is

$$N(T) \sim \frac{dT}{2\pi} \log \frac{q^{1/d} T}{2\pi e}, \quad T_{\min}^{\text{std}} = 2\pi e q^{-1/d}, \quad (2)$$

and Eqs. (1) and (2) coincide *only* at $d = 1$, where both give $2\pi e/q$.

The discriminating test. The $d > 1$ case decides between them. We computed the zeros of two Dirichlet L -functions (via Hurwitz-zeta construction and Hardy- Z sign changes) and of a genuine degree-two L -function, the Dedekind zeta of $\mathbb{Q}(i)$, $\zeta_K = \zeta \cdot L(\chi_{-4})$, conductor $q = 4$. The zeros of ζ_K are the union of the ζ zeros and the $L(\chi_{-4})$ zeros, so its counting function is exact.

L -function	d	q	framework $(2\pi e)^d/q$	standard $2\pi e q^{-1/d}$	N_{actual} at framework floor
$L(\chi_{-4})$	1	4	4.27	4.27	0 (first zero 6.02)
$L(\chi_3)$	1	3	5.69	5.69	0 (first zero 8.04)
$\zeta_{\mathbb{Q}(i)}$	2	4	72.93	8.54	51

Table 2: The degree-two case falsifies the scaling law. The framework floor sits above 51 actual zeros; the standard floor sits just after the first zero.

At $d = 2$ the framework floor $(2\pi e)^2/4 = 72.93$ sits *above* 51 *actual zeros* — flatly contradicting the framework’s own definition of $T_{\min}$ as the height below which “too few zeros exist.” Equation (1) predicts ≈ 0 zeros up to $T = 72.93$; the truth is 51, and Eq. (2) correctly predicts ≈ 50 . The standard floor 8.54 sits just after the first zero ($N = 1$), as a floor should.

Verdict (T4). The floor *value* at $d = 1, q = 1$ remains an analytic invariant. The *scaling law* $T_{\min} = (2\pi e)^d/q$ is falsified in its degree dependence: the floor scales as $2\pi e q^{-1/d}$, overshooting by a factor ~ 8.5 at $d = 2$. The ζ “survival” in earlier scoped reporting was precisely the degenerate case in which the wrong degree dependence coincides with the right one — exactly where the strip-mine anticipated the discriminator would lie.

4.2 The information-asymmetry / central-charge identity $\Delta I \equiv c$ (T4 as stated)

The program’s stated load-bearing open conjecture identifies the information asymmetry ΔI with the renormalization-group c -function. The reachable definitions are: a routing scalar $\delta_i = 0.4(\text{consistency}) + 0.3(\text{divergence}) + 0.2(\text{late_error}) + 0.1(\text{recomposition}) \in [0, 1]$, and a physical transfer-entropy asymmetry

$$\Delta I = \text{TE}(F \rightarrow G) - \text{TE}(G \rightarrow F). \quad (3)$$

We built an independent reference c -function (the Casini–Huerta entropic c -function on a free-Majorana transverse-field Ising chain, computed by a correlation-matrix method gate-validated against exact diagonalization to 10^{-13}). It recovers the Ising central charge $c = 1/2$ at criticality (fitted 0.488 at scale) and flows monotonically to 0 in the massive phase.

The identity fails (T2 $\rightarrow$ T4 as stated). The central charge is unbounded above ($c = 1$ for a free boson, $c = N$ for N free bosons, $c = 26$ for the bosonic string). Hence:

- If ΔI is the bounded routing scalar ($\in [0, 1]$), the identity is falsified for every theory with $c > 1$.
- If ΔI is the transfer-entropy asymmetry, three structural mismatches block the identity: **(i) sign** — $c \geq 0$, but a difference of transfer entropies is signed; **(ii) fixed-point value** — on a symmetric fixed point $\text{TE}(F \rightarrow G) = \text{TE}(G \rightarrow F)$ so $\Delta I = 0$, while c equals the (nonzero) central charge; **(iii) category** — transfer entropy is a temporal, directed-information-flow quantity requiring a time series, whereas c is static and spatial (Zamolodchikov; Casini–Huerta), so the two are not even defined on the same object.

The surviving reading is a theorem, not a conjecture. The only reading under which a monotone quantity “plays the role of” c is the mutual-information / entropic c -function. We confirmed numerically that this quantity is monotone non-increasing along the flow for every mass tested. But under that reading $\Delta I = c$ is not a novel conjecture: it is the Casini–Huerta entropic c -theorem, monotone by strong subadditivity. The literal transfer-entropy asymmetry cannot furnish it: it is undefined on a static, time-translation-invariant ground state and vanishes by symmetry on a symmetric bipartition.

Verdict. Every reading is accounted for: $\Delta I \equiv c$ is either false/ill-defined (transfer-entropy asymmetry, or bounded routing scalar at $c > 1$) or a restatement of an established theorem (mutual-information reading). No reading is both novel and true. The highest-collapse target is retired.

4.3 The BRA speed-superiority claim (T4 for speed; determinism survives at T1)

The BRA kernel was advanced as a faster deterministic replacement for a TensorFlow operation, with a withdrawn projection of a $489\times$ speedup. We identified the matched operation (a Gabor wave-packet render/energy, replacing $\exp(-\delta t^2/2w^2)\exp(2\pi i f \delta t)$ in f32) and benchmarked the release-mode kernel against it.

N	BRA f64 (ns/op)	TF f32 (ns/op)	ratio
16	225.6	124.0	$1.82\times$ slower
64	700.9	469.8	$1.49\times$ slower
256	2505.5	1638.0	$1.53\times$ slower
1024	8326.3	6074.5	$1.37\times$ slower

Table 3: Measured BRA vs. TF throughput. The speed-superiority claim is killed.

Verdict. BRA is $1.37\text{--}1.82\times$ *slower*; the $489\times$ claim is obliterated. One confound is recorded honestly: the comparison is BRA-f64 against TF-f32, and f64 carries $\sim 1.5\text{--}2\times$ the work, so “competitive at matched f64” is untested (plausibly near parity). The genuine surviving value is *determinism*: the integer and f64 paths are bit-identical across runs (T1). The speed claim is dead; the determinism claim — always the real one — stands.

4.4 Residual prime-orbit off-diagonal mechanisms (T2 + T3)

We tested whether any *peaked* off-diagonal mechanism contributes to the zeta spectrum beyond the diagonal prime-power terms. Using the explicit-formula dual periodogram on the zeros, the power

ratios to local baseline were: fundamentals $\log 2, 3, 5, 7$ at $\sim 3\text{--}4 \times 10^8$; prime-power harmonics $2\log 2, 3\log 2, 2\log 3, 2\log 5$ at $\sim 5 \times 10^7\text{--}4 \times 10^8$ (real peaks, suppressed by the $p^{k/2}$ weight); composites $\log 6, 10, 12, 15$ at 0.3–5.4 (baseline noise); decoys ~ 1 .

Verdict. The peaked spectrum is exactly the diagonal prime-power set $\{k \log p\}$ at $10^7\text{--}10^8$ contrast. There is a structural reason, not merely a null: the explicit-formula dual carries weight only on the von Mangoldt function $\Lambda(n)$, which is supported on prime powers, so a sum-frequency peak at $\log(pp')$ is *forbidden*. Difference-tone off-diagonal peaks were falsified in prior work. The remaining off-diagonal content is smooth (the Montgomery pair-correlation term, separately confirmed). No residual peaked off-diagonal mechanism exists; the tunnel is mapped empty.

5 Results III: class-level eliminations — the Hilbert–Pólya operator

One cannot construct and diagonalize a self-adjoint operator with spectrum equal to the zeros — that is the open problem itself — but the data forces discriminating constraints, and those constraints kill candidate *classes*.

Symmetry class (T3). The unfolded nearest-neighbour spacings of the ζ zeros match the GUE Wigner surmise (L2 distance 7.3×10^{-2}) far better than GSE (1.7×10^{-1}), GOE (2.3×10^{-1}), or Poisson (6.5×10^{-1}); the fitted level-repulsion exponent is $\beta \approx 2.12$, with variance 0.1600. Under random-matrix universality the operator must lie in the unitary class with broken time-reversal symmetry. This kills any real-symmetric / time-reversal-invariant candidate (GOE, $\beta = 1$) and any symplectic candidate (GSE, $\beta = 4$).

Berry–Keating xp (T2 + T3). The semiclassical xp count reproduces the smooth Riemann–von Mangoldt staircase to $< 0.2\%$, so it passes the smooth-density necessary condition. But xp has a continuous spectrum and no discrete eigenvalues, and the fluctuation $S(T)$ is prime-driven, which bare xp does not encode. “ xp alone is the operator” is killed as sufficient; “ xp as the smooth skeleton” survives.

Verdict. The surviving operator must simultaneously be unitary class (broken T), reproduce $(T/2\pi) \log(T/2\pi e)$, and encode the primes in its fluctuation spectrum. Two classes are mapped empty; the full construction remains open.

6 Results IV: a tension resolved by decoupling (T3 + T2)

The naive Type-I see-saw active–sterile mixing $\sin^2 2\theta \approx 4 \times 10^{-6}$ is excluded by the X-ray bound $\sin^2 2\theta < 10^{-10}$. The proposed Pati–Salam gauge-sector suppression is $(M_W/M_{W_R})^2$. Requiring it to clear the bound gives $M_{W_R} \gtrsim 16$ TeV (equivalently $v_R \gtrsim 49$ TeV), comfortably below the independent lower bounds on the Pati–Salam scale from rare decays ($K_L \rightarrow \mu e$ pushes v_R to $\gtrsim$ hundreds of TeV). So the mechanism clears the bound.

Verdict. The X-ray bound is not a live obstruction once v_R sits at its independently-required scale. But the resolution is by *decoupling*: clearing the bound requires high v_R , hence a heavy right-handed mass $M_R \sim v_R$, so the sterile state is no longer a keV-scale dark-matter candidate.

The tension is removed for the field-theory consistency reading, and the keV-sterile-dark-matter reading does not survive.

7 The accumulated dead ends

For completeness we gather the program’s prior falsifications — the empty tunnels mapped before this campaign — with the mechanism that killed each.

Killed claim	Killing mechanism
$\text{ad}^3 = 2\text{ad}$ for integers	requires $\lambda = 1/\sqrt{2}$; eigenvalue constraint fails for integers
Universal 2π inversion	$\mathfrak{sl}(4, \mathbb{R})$ adjoint is hyperbolic, not rotational; representation-specific
Wronskian bracket = Poisson bracket	Leibniz rule fails by $-fgh'$; structural claim demoted to analogy only
IR lattice imprint in zeros	z -scores null against PDG values; statistical null test
Intrinsic chirality of the bias	bias vanishes under covariant conjugation; not invariant
Signature selection via Killing form (Route A)	Cartan involution $\neq$ parity involution
Signature selection via stability (Route C)	all signatures give equal holonomy norms
Coleman–Weinberg $6 \rightarrow 5$ parameter reduction	$\tan \beta$ is a gauge-protected flat direction (RG-invariant ratio, multiplicative two-loop QCD, β -independent thresholds)
α_2 optional	forbidden by representation theory, $T^A \Phi = 0$
β_c explanatory	tree-level no-go from mass-matrix rank
Yukawa rescue of $\tan \beta$	impossible via the identity $M_u - M_d = (h - \tilde{h})(\kappa_1 - \kappa_2)$
Action-proximity (τ) off-diagonal mechanism	inconsistent with the incommensurable surrogate null
Difference-tone / Tartini off-diagonal mechanism	inconsistent with the surrogate null
“3-6-9” attractor	retired as falsified (no structural support)

Table 4: Prior dead ends gathered from the program. Each is a first-class negative result with an explicit killing mechanism.

8 Discussion: recurring patterns

Relations survive; bare numbers are refractions. The single most consistent finding (Sec. 3) is that the quantities surviving a change of instrument are relations, ratios, and equalities, while convention- or scale-laden numbers are not. This is precisely what one expects of physics, whose observables are relations; it also gives a fast triage heuristic for which claims to suspect.

The degenerate-case trap. The height-floor law (Sec. 4.1) survived every test available at $d = 1$ because at $d = 1$ a wrong degree dependence coincides exactly with the right one. A claim can pass indefinitely on the degenerate slice of its parameter space; the discriminating test lives off that slice, and naming where it lives *before* testing is what turned a scoped survival into a clean falsification.

Category mismatch. The $\Delta I \equiv c$ identity (Sec. 4.2) failed most fundamentally on a category error: a temporal, directed quantity identified with a static, spatial one. When an identity is proposed between objects from different categories, the burden is a bridge construction, not a numerical coincidence.

Necessary but insufficient. The Berry–Keating xp operator (Sec. 5) passes a necessary condition (the smooth count) and is nonetheless insufficient (no discrete spectrum, no prime fluctuations). Passing a necessary test is not survival; a candidate must clear every condition the data forces.

Methodological self-corrections. The campaign caught and discarded several of its own artifacts before they could be read as signal: a band-relative-error onset statistic dominated by low-height discreteness; a sub-sample indexing bug in a deviation metric; a “broken-lens” control that failed to break the GUE statistic for an identifiable reason; projected speedups mistaken for measurements; and a SIGPIPE harness artifact that reported a passing test as a failure. Each was recorded, not hidden. The discipline that catches these is the same one that produces the kills: pre-registration, negative controls, and the refusal to let analysis pose as measurement.

9 What survives, and what remains open

Survivors (earned by elimination). The dimensionless ratio $\tilde{h}/h = 2/3$ (RG-invariant); the coupling-unification equality $g_A = g_L = g_R$; the analytic floor $2\pi e$ at $d = 1, q = 1$; the Koide-geometric origin of λ_ϕ ; the GUE statistics and the prime-power structure $\{k \log p\}$ of the zeta spectrum (the latter forced by Λ -support). These survive not because they were proven from first principles here, but because the alternatives around them were eliminated.

Open. The Hilbert–Pólya operator construction (constrained but not constructed); the action principle for the grading functional; the extension of the spectral results to general L -functions (now with a corrected — standard — height-floor scaling); and the conceptual question of whether a static, positive, canonically-normalized information quantity exists that could non-trivially realize a c -function role.

10 Conclusion

The campaign retired the two highest-collapse claims of the program — the $\Delta I \equiv c$ identity and the $T_{\min} = (2\pi e)^d/q$ scaling law — and both were the program’s own. Several “forced” constants were reclassified as refractions, with their underlying relations surviving. A speed claim was killed and a determinism claim confirmed. Two operator classes were eliminated and a peaked-spectrum tunnel mapped empty. None of these required a new positive theory; each followed from making destruction cheap, pre-registering the kill condition, and treating a clean negative as a result worth the same care as a proof. The survivors are credible in proportion to what was removed around them. That is the whole of the method: do not prove the gold is there; remove everything that is not gold, and keep an honest ledger of every tunnel.

A Reproducibility

All numerical tests are reproducible from the accompanying scripts, *with the caveat that the two scripts named below were run on external session machines and are not committed to the public*

repository: the recorded verdicts rest on the logged outputs quoted here and in `docs/Elimination_Ledger.md`. The degree-two falsification (Sec. 4.1) uses `q3_scaling_test.py` (mpmath Dirichlet- L zeros via Hurwitz-zeta construction and Hardy- Z sign changes; ζ zeros from the first 10^5 accurate to 3×10^{-9} ; the degree-two case built as $\zeta \cdot L(\chi_{-4})$). The reference c -function (Sec. 4.2) uses `q4_cfunction_testbed.py` (free-fermion BdG correlation-matrix method, gate-validated against exact diagonalization). Verdicts and tiers are recorded in this paper (the ledger of record for the campaign) and mirrored in the repository’s living kill log, `docs/Elimination_Ledger.md`. Random seed 20260423 where randomness appears.

Amendment: scope and evidence, 24 September 2026

The September carrier audit adds finite obstructions to the existing elimination discipline: the cover shear fails deck commutation, the balanced triplet is not a singlet, the capacitor normalization differs by a factor of two, and the stated torsion term is algebraic. The action audit also disproves uniqueness from a neutral restriction by a stable/flat/unstable counterfamily. These retire specific implications under their assumptions. They neither eliminate all future ACS models nor prove the remaining conjectures.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository’s `papers/updates/` directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.